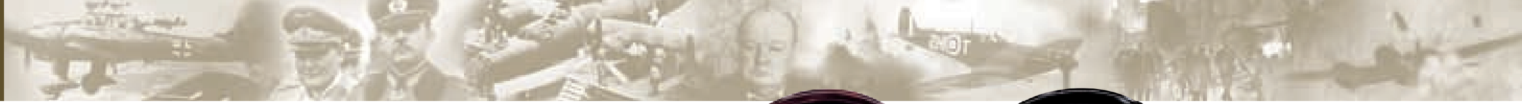

OPERATIONS ROOM
Members of the British Women's Auxiliary Air Force push wooden blocks and arrows representing enemy bombers around a map as information from Observer Corps Centres comes through on their headsets. Above them, controllers watch the progress of the enemy.

SCRAMBLE FOR THE SKIES
This photograph (taken in Duxford, England, before WWII) is of a demonstration given for the press by the RAF. The purpose of the exercise was to show the speed at which the airmen could reach their aircraft when called upon.

While the German commanders were confused in their objectives, the British were focused and organized to a single end. Fighter Command chief Hugh Dowding recognized that for the RAF, surviving as a coherent defensive force was enough to constitute victory. By avoiding committing too large a part of his resources to the combat prematurely, and making his fighters concentrate on knocking out German bombers, Dowding conducted a ruthless and calculating campaign of attrition.

CODED CLOCKS
Clocks in operations rooms had color-coded segments. Arrows placed on map tables used the same color coding, showing when their position was last updated.

British defenses

The British air-defense system was the most sophisticated in the world, a triumph of organization and applied technology. In its front line were the radar stations that identified Luftwaffe aircraft approaching Britain's shores. Radar operators provided a dense flow of raw data that was processed at a centralized "filter room" and forwarded to the people who controlled operations. In the operations rooms members of the Women's Auxiliary Air Force converted the information into 3-D graphic form by pushing wooden blocks around on a map with croupier's rakes, watched from a balcony above by the controllers. The controllers at the headquarters of the four fighter groups – each responsible for the defense of an area of the country – decided when, where, and at what strength aircraft should be sent up to meet the Luftwaffe. Controllers at sector level, with typically three or four squadrons under them, were responsible for directing the fighters on to their targets by radioed instructions.

This system was fallible. The problem of distinguishing between friendly and enemy aircraft was tackled by equipping RAF fighters flying in a radar zone with IFF (Identification Friend or Foe) devices, which gave them a distinctive radar signature, but there were often not enough IFF sets to go around, and they did not always work. Also, the Bf 110s of the Luftwaffe's Erprobungsgruppe 210 discovered that by flying close to ground level they could creep under the radar unobserved. But the